

Constructing and Solving Equations — Paper B

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every equation on this paper has a whole-number solution unless it says otherwise.
- Whatever you do to one side, do to the other. Write the operation beside each line, like “ -4 ” or “ $\div 3$ ”.
- Strip the numbers off the x first and divide last. Dividing early makes you divide every term.
- Every answer can be checked by putting it back into the original equation. Do it every time — it costs ten seconds and it catches nearly every error.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – ONE STEP, THEN TWO

- 1.** Complete the table. Each empty box is worth one mark. In the last column, show the substitution that proves your answer is right. [6]

Equation	Solution	Check (substitute back)
$x - 12 = 5$		
$8x = 72$		
$2x + 7 = 19$		

Write the operation beside every line you produce. Seeing it written twice is what stops one-sided working.

- 2.** Solve:

(a) $x + 14 = 9$ [1]

(b) $6x = 54$ [1]

(c) $x \div 3 = 12$ [1]

(d) $9 - x = 4$

[1]

3. Solve:

(a) $3x + 11 = 26$

[1]

(b) $5x - 7 = 33$

[1]

(c) $2x + 15 = 7$

[1]

(d) $4x - 5 = 23$

[1]

SECTION 2 – BOTH SIDES, AND BRACKETS

4. Solve, showing each step:

(a) $6x - 5 = 2x + 15$

[2]

(b) $8x + 3 = 5x + 21$

[2]

5. Solve:

(a) $4(x + 2) = 28$

[2]

(b) $2(3x - 1) = 22$

[2]

SECTION 3 – BUILDING THE EQUATION YOURSELF

6. I think of a number, multiply it by 6, then add 5. The result is 41.

(a) Using n for the number, form an equation.

[1]

(b) Solve it.

[1]

(c) Check your answer against the original instructions.

[1]

7. A triangle has sides of length x cm, $(x + 2)$ cm and $(x + 4)$ cm. Its perimeter is 30 cm.

(a) Form an equation in x .

[1]

(b) Solve it.

[1]

(c) Write down the three side lengths and check they add to 30. [1]

8. A father is 4 times as old as his son. Together their ages total 45.

(a) Using s for the son's age, form an equation. [1]

(b) Solve it, and state both ages. [1]

End of paper. Check every answer has a unit or a form the question actually asked for.